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Abstract

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1 Intra-Dataset Exploration and Visualization

Exploratory Data Analysis (EDA) represents a fundamental step in DNA methylation studies, particularly in cancer epigenomics, where alterations in global methylation levels, sample-level variability, and locus-specific instability often precede detectable phenotypic changes. Before applying any preprocessing, filtering, bias correction, or machine-learning models, each dataset must be evaluated independently in order to assess its internal structure, detect technical anomalies, and identify biologically meaningful patterns.

In the context of breast cancer, epigenetic deregulation manifests through both global and local phenomena: genome-wide hypomethylation, focal hypermethylation of tumor-suppressor regions, and the presence of pre-neoplastic “field defects” in histologically normal tissues close to tumors [1], [2], [3]. The primary biological goal of this thesis—to *identify DNA methylation alterations in normal tissues and evaluate whether they may contribute to the transition toward tumorigenesis*—motivates this detailed intra-dataset investigation. This section analyzes each dataset (GSE69914 [1.1], GSE225845 [1.2], GSE287331 [1.3]) separately and systematically, focusing on: missingness and quality control, global methylation distributions, sample-level summaries, CpG-level instability, correlation structure, dimensionality reduction, and genome-context coverage.

1.1 Dataset GSE69914

Dataset overview The GSE69914 dataset provides genome-wide DNA methylation profiles generated using the Illumina Infinium HumanMethylation450 BeadChip on bisulfite-converted DNA, covering more than 480,000 CpG sites. The series, contributed by Teschendorff and Widschwendter and publicly released on June 18, 2015, is available on GEO under accession [GSE69914](#).

In total, the dataset includes **407 breast tissue samples** spanning the three tissue states central to this thesis:

- 50 Normal samples, representing the physiological methylation baseline;
- 42 Tumor-adjacent samples, reflecting early epigenetic alterations in proximity to the tumor;

- 305 Tumor samples, providing the malignant endpoint of the Normal $\rightarrow$ Adjacent $\rightarrow$ Tumor axis.

A small subset consists of BRCA1-related samples (7 normal carriers and 3 tumors). BRCA1 encodes a key DNA-repair protein, and pathogenic variants substantially increase lifetime breast and ovarian cancer risk [4], adding a hereditary-risk dimension to the cohort. Since the primary objective of this thesis is to characterise epigenetic differences along the Normal-Adjacent-Tumor axis that are not driven by strong germline predisposition, BRCA1-mutated samples were excluded from the core exploration and preprocessing pipeline. Their inclusion would introduce a distinct hereditary subtype with potentially different baseline methylation patterns, confounding the interpretation of field-effect alterations in sporadic breast cancer. The final working matrix therefore contains 397 tissue samples and 485,512 CpG probes, offering a structured and biologically coherent dataset for intra-dataset exploratory analyses.

Missingness and data quality An assessment of missing values across all samples and CpG probes confirmed that the GSE69914 methylation matrix contains *no missing entries* of any kind. The dataset is therefore complete, technically clean, and well suited for direct exploratory analysis and visualization without requiring imputation or preliminary data recovery steps.

Global methylation distributions To characterise the overall methylation landscape of the dataset, I first examined the distribution of β -values (fractional methylation in $[0, 1]$). The group-wise mean density curve (Fig. 1) displays the characteristic *bimodal* profile of Illumina 450k arrays, with peaks near unmethylated ($\beta \approx 0$) and fully methylated ($\beta \approx 1$) CpG sites. This pattern reflects the underlying biology of CpG regulation, where many loci tend to be either transcriptionally active (hypomethylated) or repressed (hypermethylated) [2], [5].

When comparing tissue groups, a clear gradient emerges. Tumor samples show a slightly flatter high- β peak and a broader low- β tail, consistent with the well-described phenomenon of **global hypomethylation** and increased heterogeneity in cancer [1]. Tumor-adjacent samples lie between Normal and Tumor, suggesting early epigenetic drift and subtle field effects occurring in histologically non-neoplastic tissue [3]. Normal samples exhibit the sharpest and most stable bimodality, representing the expected physiological baseline.

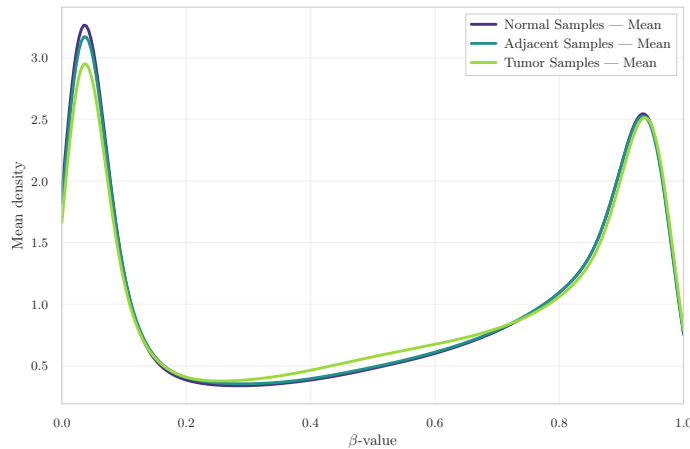


Figure 1: Group-wise mean β -value density.

CpG-level instability and recurrent outliers To characterise locus-specific instability, I examined CpG sites that show the strongest deviations across samples. The heatmap of the top outlier loci (Fig. 2a) indicates that epigenetic disruption is *not uniform*: specific CpG sites and specific samples display extreme deviations, rather than a diffuse genome-wide shift. Moreover, both directions of alteration are present — focal **hypermethylation** (very high β) and focal **hypomethylation** (very low β). In cancer biology, hypermethylation can silence tumor-suppressor regions, whereas hypomethylation can derepress oncogenic pathways and weaken genomic stability, reflecting the classic “too much and too little methylation” behaviour of tumor genomes [1].

Complementary $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Fig. 2b) show a clear shift toward hypomethylation in Tumor samples and a subtler but detectable drift in Adjacent tissues. This provides evidence that epigenetic alterations emerge early in histologically normal tissue located near the tumor, supporting the field-cancerization model [3].

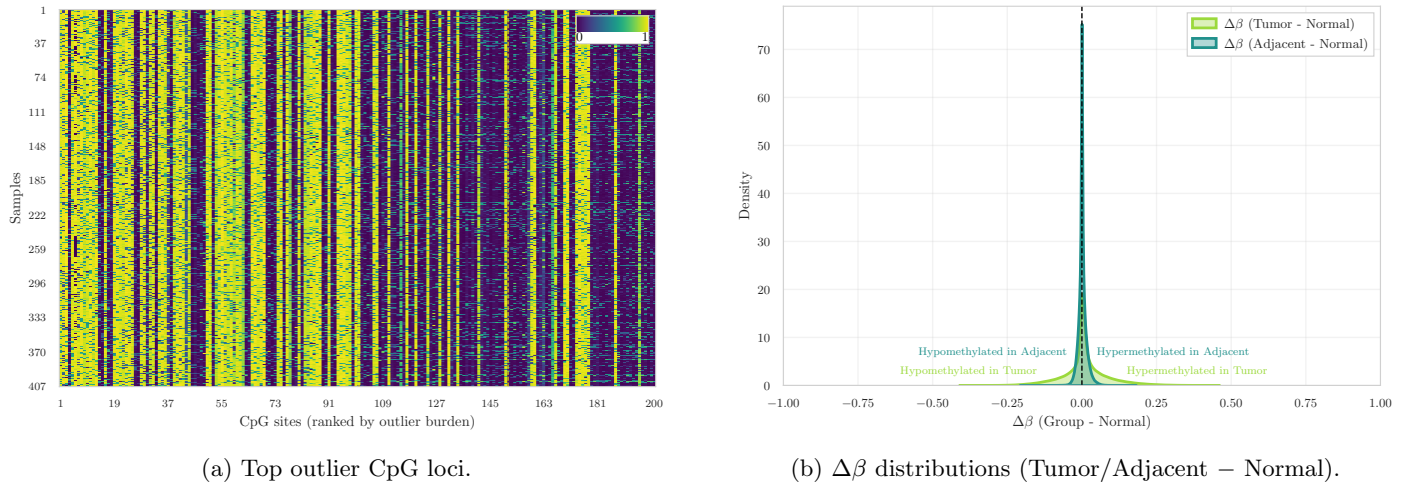


Figure 2: CpG-level instability: heatmap of top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right).

Variance and correlation structure To quantify within-group epigenetic variability, I examined the distribution of CpG-wise variance across samples. The density curves (Fig. 3a) show that Tumor samples have markedly higher variance, reflecting increased epigenetic instability. In contrast, Normal and Tumor-adjacent samples display almost identical variance distributions that are tightly concentrated near zero. This indicates that, at the level of CpG-wise variability, Adjacent tissues do not yet exhibit detectable divergence from Normal samples, despite being histologically proximate to the tumor.

A complementary perspective is provided by the sample correlation heatmap (Fig. 3b), which shows the expected high level of pairwise similarity across all samples. This behaviour is typical of genome-wide DNA methylation data, where a large fraction of CpG sites is stable across individuals, resulting in consistently strong correlations [6].

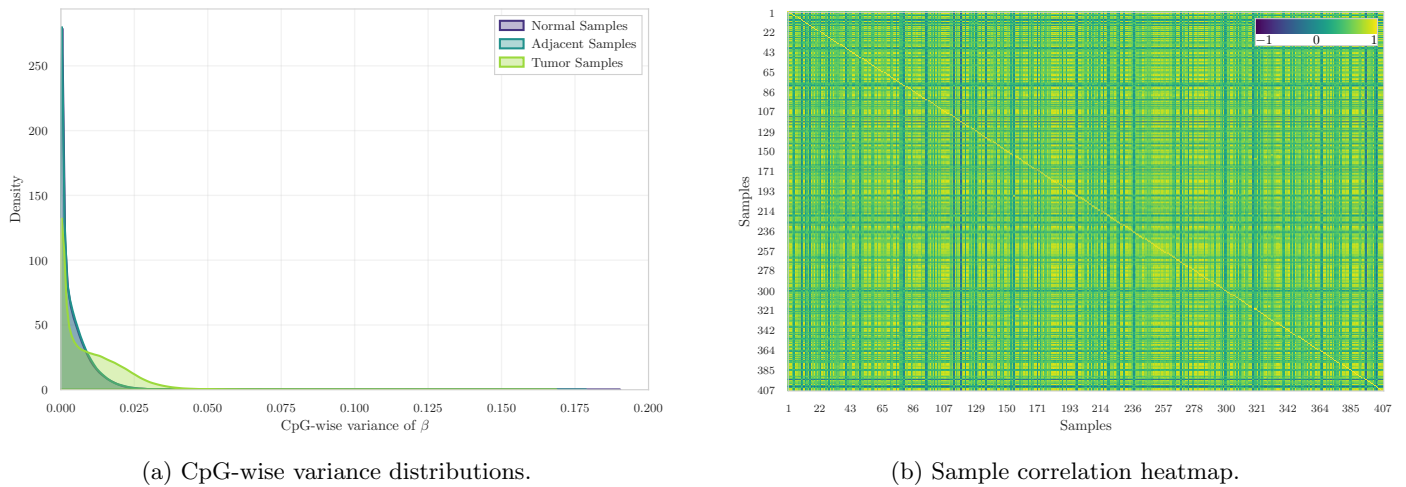


Figure 3: Variance and correlation structure across Normal, Adjacent, and Tumor samples.

Low-dimensional embeddings Dimensionality reduction was applied using PCA, t-SNE, and UMAP to visualise global similarities among samples (Fig. 4). Across all three methods, Normal and Tumor-adjacent samples show substantial overlap, with no clear separation between these two groups. Tumor samples appear more dispersed, but only mildly shifted relative to the Normal/Adjacent cluster, and no sharp cluster boundaries emerge in any embedding.

Overall, these projections indicate that global methylation patterns alone do not provide strong discriminative structure among the three tissue states. This suggests that group differences are subtle at the whole-methylome level and are better captured through locus-specific analyses (e.g., outlier burden or $\Delta\beta$), rather than through global unsupervised embeddings.

Genome annotation coverage CpG probes were annotated using the official *Infinium MethylationEPIC v1.0 B5* [7] manifest file, the standard Illumina resource containing genomic coordinates, probe type, CpG island context (Island, Shore, Shelf, Open Sea), and gene-level annotations. Using this manifest, the GSE69914 dataset shows the expected

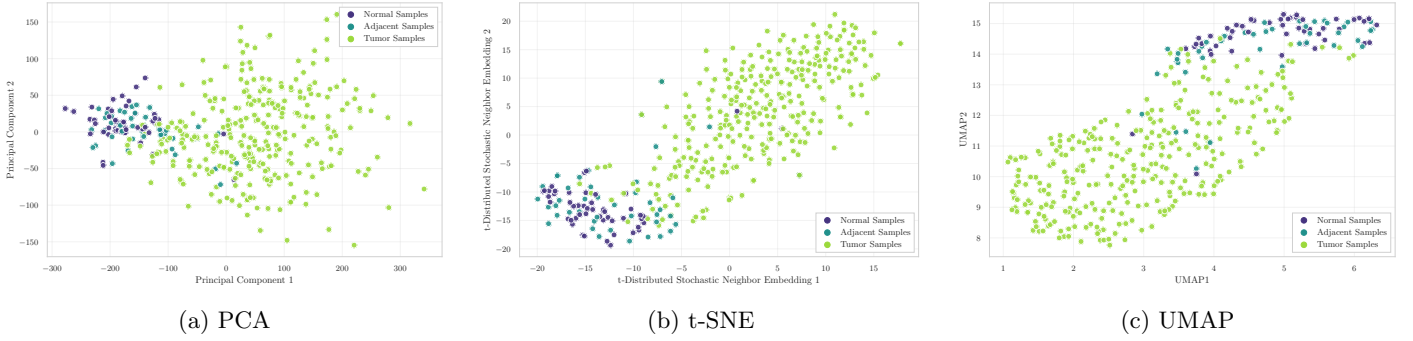


Figure 4: Low-dimensional embeddings of GSE69914 samples.

non-uniform genomic distribution of CpG contexts: the largest fraction of probes maps to open-sea or non-island regions, followed by CpG islands, then north and south shores, and finally north and south shelves. This ordering mirrors the characteristic architecture of the EPIC/HM450 arrays reported in the literature. Zhou et al. [8] describe the EPIC design and its inheritance of the HM450 probe structure, while Pidsley et al. [9] document the enrichment of probes in CpG islands and shores with decreasing density in shelves and open-sea regions. The coverage observed in GSE69914 is therefore consistent with the known genomic composition of the Illumina methylation arrays, supporting correct manifest alignment.

Dynamic-network proxies Although this analysis does not follow the full mathematical formalism of Dynamic Network Biomarkers (DNBs), the degree-based and network-level representations in Fig. 5 provide a qualitative view of how methylation perturbations may organize at the network level. DNB theory, originally formulated for gene-expression and regulatory networks, predicts that near a critical transition a subset of components displays increased internal fluctuations and coordinated behaviour [10], [11], [12]. Here I adapt this conceptual framework to CpG-level methylation patterns by examining the structure of CpG-gene degree distributions and the organization of $\Delta\beta$ -driven network layouts. This allows me to explore whether tumour-associated instability manifests as locally coordinated methylation shifts at the network level, even without implementing the complete DNB pipeline.

The degree-rank plot (Fig. 5a) displays a heavy-tailed distribution of CpG counts per gene, consistent with the heterogeneous hub-periphery structure typically observed in biological systems. In the Tumor vs Normal comparison (Fig. 5b), a compact nucleus of nodes with moderately high degree emerges, accompanied by larger $\Delta\beta$ deviations. This pattern suggests a more perturbed and locally coordinated subset of CpG/gene units in tumor tissue, qualitatively consistent with increased network instability. In contrast, the Adjacent vs Normal network (Fig. 5c) lacks a comparable hub-like structure and shows smaller, more diffuse deviations, indicating that the adjacent tissue may exhibit only early or weakly organized perturbations.

Although these observations do not satisfy the full mathematical criteria for identifying a DNB module, they align conceptually with the idea that tumor tissue occupies a more unstable network regime, whereas adjacent tissue remains closer to the stable (pre-transition) configuration.

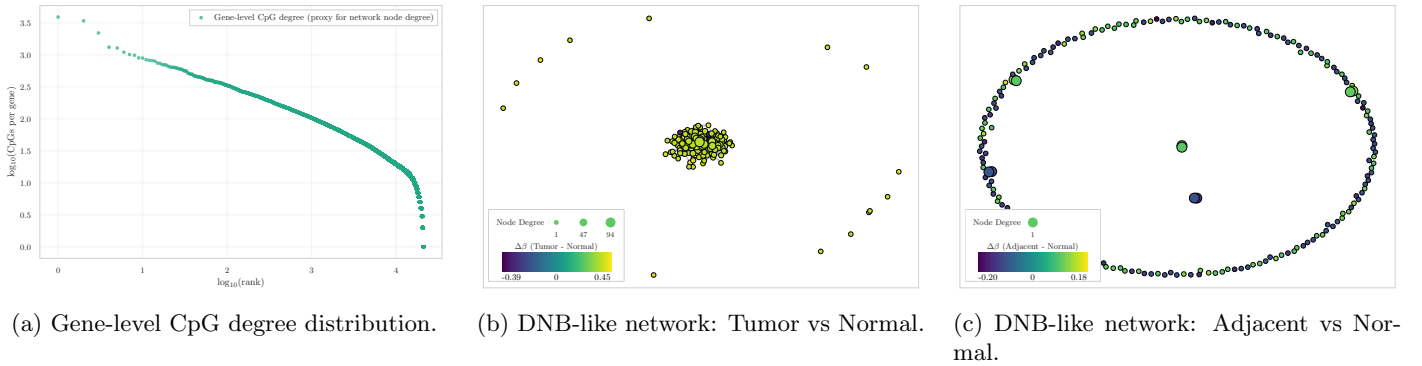


Figure 5: Dynamic-network proxy visualizations for GSE69914.

Overall interpretation The exploratory analysis of GSE69914 reveals a consistent but overall subtle separation between the three tissue states. Normal and Tumor-adjacent samples share highly similar global methylation profiles,

variance distributions, correlation structure, and low-dimensional embeddings, indicating that large-scale methylome organisation remains largely conserved in adjacent tissue. Tumor samples show the expected increase in heterogeneity and a shift toward hypomethylation, but these differences remain modest at the whole-methylome level and do not generate clear separation in unsupervised embeddings.

Locus-specific analyses, however, highlight focused patterns of disruption. Outlier CpGs show both hyper- and hypomethylation events, and $\Delta\beta$ distributions reveal a distinct hypomethylation bias in Tumor samples and a milder drift in Adjacent tissue. Dynamic-network proxies further suggest a more compact and perturbed core in Tumor samples, whereas Adjacent tissue remains more diffuse and weakly perturbed.

Altogether, GSE69914 presents a technically clean dataset with well-structured methylation patterns and biologically interpretable deviations. The results indicate that tumour-related alterations emerge predominantly at focal CpG loci rather than at the global methylome level, and that Adjacent tissue reflects early but subtle perturbations consistent with field-effect phenomena.

1.2 Dataset GSE225845

Dataset overview The GSE225845 dataset provides genome-wide DNA methylation profiles generated using the Illumina Infinium MethylationEPIC BeadChip (850K) on bisulfite-converted DNA, covering more than 850,000 CpG sites. The series, part of the NCI-Maryland Breast Cancer Cohort and publicly released on October 15, 2023, is available on GEO under accession [GSE225845](#).

In its original design, the study profiled **402 fresh-frozen breast tissues from 289 women**, including 185 tumor samples, 113 paired adjacent normal samples, and 104 normal tissues from reduction mammoplasty. Neighbourhood-level socioeconomic deprivation was quantified via a Neighborhood Deprivation Index (NDI) and linked to DNA methylation and gene-expression changes in cancer- and immune-related pathways.

For the purposes of this thesis, I focus on the three tissue states central to the Normal $\rightarrow$ Adjacent $\rightarrow$ Tumor axis using the harmonized working matrix. After quality control and label harmonization, this matrix includes **477 breast tissue samples**:

- 113 Normal samples, representing the physiological methylation baseline;
- 140 Tumor-adjacent samples, reflecting early epigenetic alterations in proximity to the tumor;
- 224 Tumor samples, providing the malignant endpoint of the Normal $\rightarrow$ Adjacent $\rightarrow$ Tumor axis.

The dataset also includes detailed metadata on demographic and clinical covariates (e.g., age at surgery, race, sex, tissue descriptors).

The final working methylation matrix used in the exploratory analyses therefore contains **477 tissue samples and 750,426 CpG probes**, providing a high-resolution EPIC-based dataset for intra-dataset exploratory analyses.

Missingness and data quality Missingness assessment shows a fully complete matrix: both samples and CpG probes have zero median missingness, and no entries exceed standard QC thresholds. Accordingly, the dataset requires *no imputation or corrective preprocessing* at this stage and is technically well suited for direct exploratory analysis.

Global methylation distributions The group-wise mean β -value densities (Fig. 6) exhibit the characteristic bimodal profile of EPIC arrays, with peaks near unmethylated ($\beta \approx 0$) and fully methylated ($\beta \approx 1$) CpG sites. Across the three tissue groups, the curves are nearly superimposed, indicating highly similar global methylation landscapes.

Normal and Adjacent samples show essentially identical profiles, while Tumor samples display only minimal deviations—slightly higher density at low β values and a very subtle decrease in the high- β peak. Overall, the dataset appears globally homogeneous, with group differences likely emerging at more localized, CpG-specific scales rather than at the whole-methylome level.

CpG-level instability and recurrent outliers To characterise locus-specific instability, I examined the 200 CpG sites with the highest outlier burden across samples. The heatmap of these loci (Fig. 7a) shows that instability in this subset is highly *directional*: almost all recurrent outlier events correspond to very low β values, indicating focal **hypomethylation**. In other words, within the top outlier CpGs, the dominant perturbation pattern is a consistent loss of methylation across many samples. This reflects instability concentrated on a small set of CpGs rather than a diffuse genome-wide effect.

The $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Fig. 7b) are sharply peaked around zero but display a clearly heavier tail for $\Delta\beta < 0$ than for $\Delta\beta > 0$. Tumor samples show the longest negative tail, consistent with more pronounced hypomethylation, while Adjacent tissues exhibit a milder yet still detectable excess of hypomethylated loci. This pattern emphasizes the “too little methylation” component of cancer-associated epigenetic change [1] and suggests that adjacent tissue already shares, in attenuated form, the same hypomethylation bias observed in tumors, in line with a field-effect model [3].

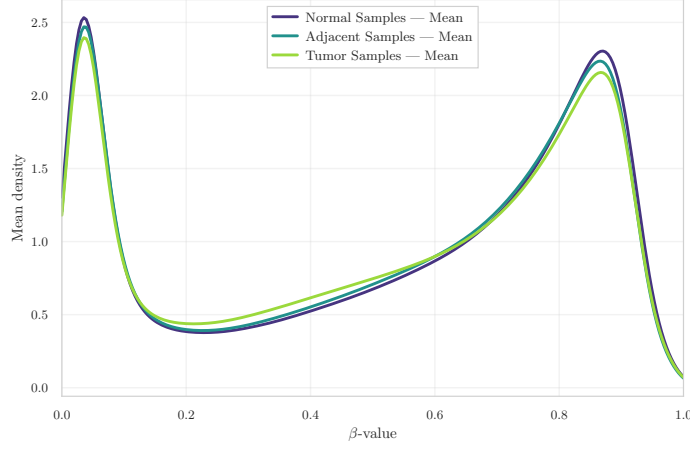
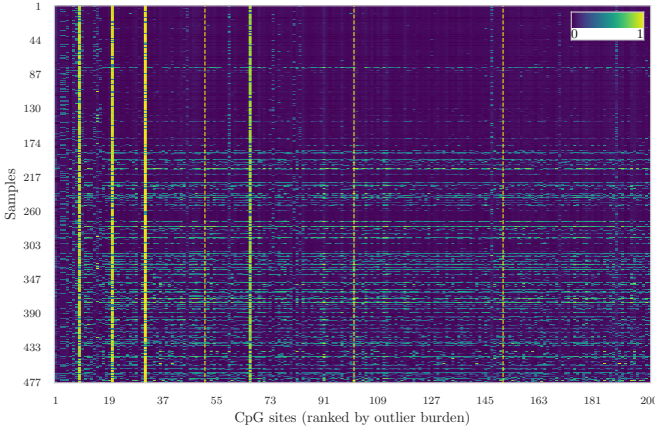
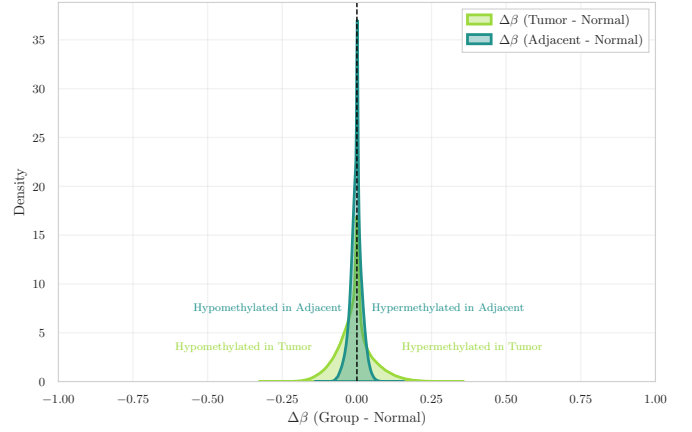


Figure 6: Group-wise mean β -value density.



(a) Top outlier CpG loci.



(b) $\Delta\beta$ distributions (Tumor/Adjacent – Normal).

Figure 7: CpG-level instability: heatmap of top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right).

Variance and correlation structure The CpG-wise variance distributions (Fig. 8a) show that the three groups exhibit distinct levels of variability. Normal samples display the lowest and most concentrated variances, whereas Tumor samples present a clearly longer tail, indicating a subset of CpGs with higher instability. Adjacent samples differ from Normal but do not follow a simple intermediate pattern between Normal and Tumor; rather, their variability profile is partially overlapping with both groups, without a clear ordering along a progression axis.

The sample-sample correlation heatmap (Fig. 8b) shows that correlations are almost entirely positive across the dataset. Most values fall within a narrow positive range, indicating a globally homogeneous methylation background with only mild fluctuations between samples. No strong blocks of high or low correlation are visible, suggesting that differences arise from localized CpG-specific variation rather than large-scale shifts in the global methylation structure.

Low-dimensional embeddings Dimensionality reduction was applied using PCA, t-SNE, and UMAP to visualise the global organisation of the samples. The PCA (Fig. 9a) plot shows a strong overall overlap among the three groups, with only a mild elongation of the Tumor samples along the second principal component. No clear separation emerges at this linear scale.

Nonlinear embeddings reveal more structure. In t-SNE space (Fig. 9b), Normal and Adjacent samples form partially separated regions, although both groups contain scattered outliers. Tumor samples are markedly more dispersed and extend over a wider area, reflecting higher global heterogeneity; several Tumor samples appear as isolated points far from the main cloud.

The UMAP (Fig. 9c) embedding displays a similar pattern: Normal and Adjacent samples are relatively compact, while Tumor samples occupy a broad and diffuse region with many peripheral outliers. Despite these differences in spread, none of the methods produces sharply defined or clearly isolated clusters.

Overall, the embeddings suggest that group differences do not manifest as distinct global subspaces but rather as

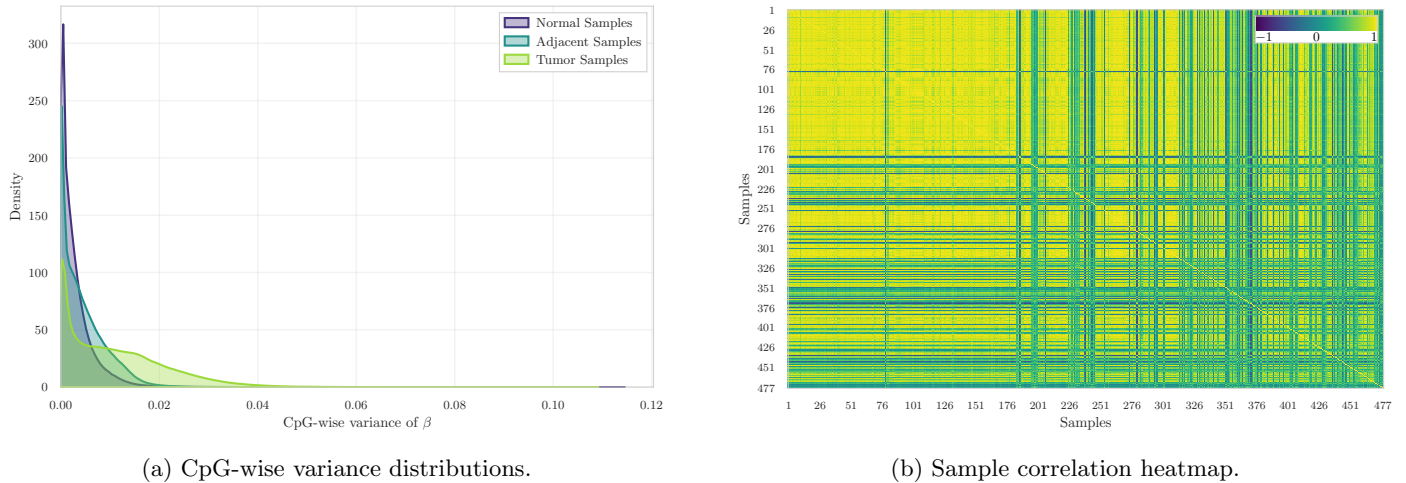


Figure 8: Variance and correlation structure across Normal, Adjacent, and Tumor samples.

differences in dispersion and local structure, consistent with heterogeneity being more prominent than large-scale shifts in the whole-methylome profile.

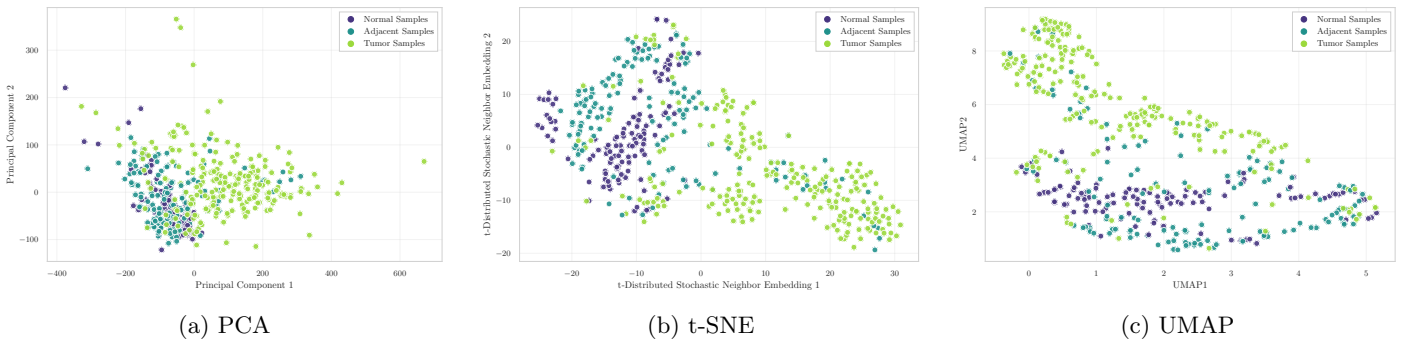


Figure 9: Low-dimensional embeddings of GSE225845 samples.

Genome annotation coverage CpG probes were annotated using the official *Infinium MethylationEPIC v1.0 B5* manifest [7]. As expected for the EPIC platform, the genomic contexts are strongly unbalanced. The majority of probes fall into the non-annotated (open-sea or non-island) category, which represents the largest single class. CpG islands constitute the second most abundant category, followed by north and south shores in comparable proportions. North and south shelves account for the smallest fractions.

This distribution mirrors the characteristic architecture of Illumina methylation arrays, where open-sea regions dominate the probe set, islands and shores form intermediate classes, and shelves are only sparsely represented. The annotation profile observed for GSE225845 is therefore fully consistent with the expected genomic composition of the EPIC array and confirms correct alignment with the manifest.

Dynamic-network proxies Although not implementing the full Dynamic Network Biomarker (DNB) framework, the degree-based summaries and the $\Delta\beta$ -driven network layouts provide a qualitative view of how local methylation perturbations organise in this dataset.

The degree-rank distribution shows a characteristic heavy-tailed shape, with a small number of genes carrying many annotated CpGs and a long tail of low-degree nodes. This pattern is typical of heterogeneous biological systems where a few elements act as local hubs.

In the Tumor vs Normal layout, nodes form a compact and cohesive central cluster, with only a limited number of isolated points at the periphery. This indicates that the strongest Tumor–Normal differences tend to concentrate on a relatively coherent subset of CpG–gene units.

The Adjacent vs Normal layout displays a markedly different organisation: a few small clusters appear near the centre, but the majority of nodes are scattered across the embedding, forming a broad ring-like structure. This suggests weaker

and less coordinated perturbations, with many CpG–gene units showing small, dispersed deviations rather than a shared local pattern.

Overall, these proxy visualisations suggest that Tumor–Normal contrasts produce a more compact and locally coherent signal, whereas Adjacent–Normal differences are more diffuse and fragmented.

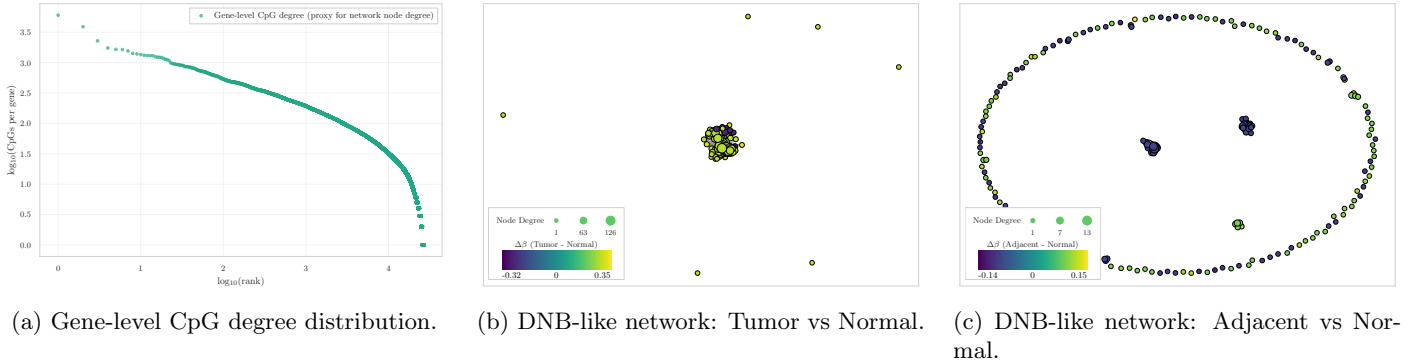


Figure 10: Dynamic-network proxy visualizations for GSE225845.

Overall interpretation The exploratory analysis of GSE225845 reveals that, at a global scale, the three tissue states share highly similar methylation profiles. Mean β -value densities are almost superimposed across groups, and both sample–sample correlations and PCA projections show a substantial degree of overlap. These results indicate that large-scale methylome structure is broadly conserved across Normal, Adjacent, and Tumor tissues, with group differences not manifesting as broad genome-wide shifts.

Local analyses, however, reveal clearer distinctions. Among the 200 CpGs with the highest outlier burden, instability is strongly directional: the dominant pattern is focal hypomethylation, especially in Tumor samples, with Adjacent tissues showing a milder but consistent excess of negative $\Delta\beta$ values. Variance distributions further support this picture, with Tumor samples exhibiting a heavier tail and Adjacent tissues differing from Normal without forming a simple intermediate pattern.

Nonlinear embeddings (t-SNE and UMAP) highlight differences in dispersion rather than cluster separation: Normal and Adjacent samples form relatively compact regions with some outliers, whereas Tumor samples are more widely spread, reflecting increased heterogeneity. Finally, network-based proxies show that Tumor–Normal contrasts produce a compact central nucleus of perturbed CpG–gene units, while Adjacent–Normal differences appear more fragmented and diffuse.

Overall, GSE225845 emerges as a technically clean and globally homogeneous dataset in which tissue-state differences are subtle at the whole-methylome level but become more apparent when focusing on localized CpG-specific perturbations, dispersion patterns, and small-scale network structure.

1.3 Dataset GSE287331

Dataset overview DA RISCRIVERE!

The GSE287331 dataset provides genome-wide DNA methylation profiles generated using the Illumina Infinium MethylationEPIC v1.0 BeadChip on bisulfite-converted breast tissue DNA, covering more than 700,000 CpG sites (702,573 CpGs in the working matrix). The series, contributed by Dennis *et al.* and publicly released on June 18, 2025, is available on GEO under accession [GSE287331](#).

The original study is organised along a “tumor proximity axis” (TPxA), spanning healthy donated breast tissue from cancer-unaffected donors (HDB), contralateral unaffected breast (CUB), ipsilateral opposite quadrant (OQ), tumor-adjacent normal tissue (AN), and tumor (TU) samples. In this thesis, these five tissue categories are mapped to a consistent three-class scheme (Normal, Adjacent, Tumor) to enable cross-dataset comparison.

The final working methylation matrix used for intra-dataset exploration contains **446 breast tissue samples**:

- 182 Normal samples, corresponding to healthy donated breast (HDB) tissue from cancer-unaffected donors and representing the physiological methylation baseline;
- 195 Adjacent samples, aggregating benign tissues along the tumor proximity axis (CUB, OQ, AN) and capturing pre-neoplastic or field-defect contexts;
- 69 Tumor samples (TU), providing the malignant endpoint of the HDB $\rightarrow$ CUB $\rightarrow$ OQ $\rightarrow$ AN $\rightarrow$ TU axis.

All methylation levels are stored as β -values in $[0, 1]$, yielding a compact working matrix of 446 tissue samples by 702,573 CpG probes that is well suited for large-scale exploratory analysis and downstream modelling.

Evidence of Missing Infinium Type I/II Probe Bias Correction During the exploratory analysis performed uniformly across all three datasets, the per-sample mean β distributions (Fig. 11a, Fig. 11b, Fig. 11c) revealed a marked inconsistency in GSE287331. Whereas GSE69914 and GSE225845 displayed a single broad peak of global methylation levels, GSE287331 showed a much narrower distribution, shifted towards higher β -values and characterized by two distinct modes in the high- β range. This unusual right-shifted bimodality, absent in the other datasets, suggested that a key preprocessing step—namely the correction of the Infinium Type I/II probe-design bias—might not have been applied in the processed data released by the authors.

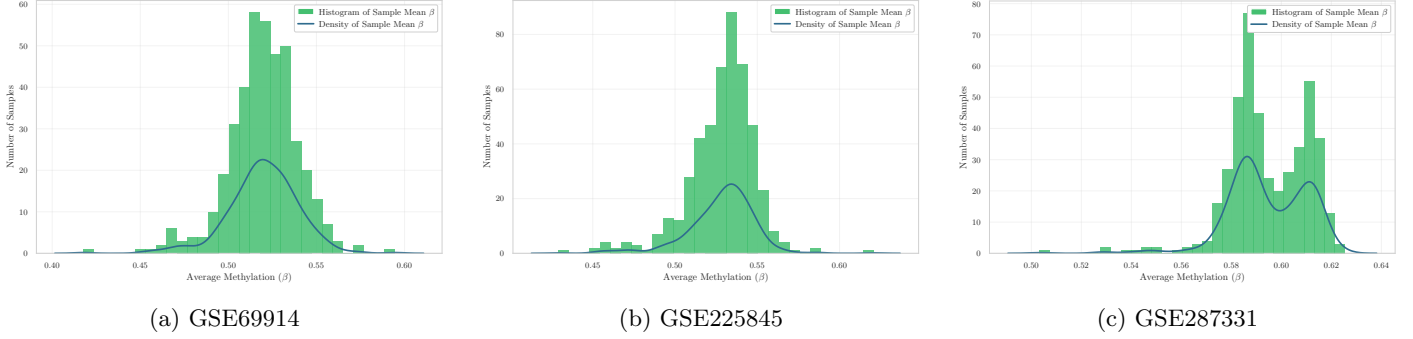


Figure 11: Per-sample mean β distributions for the three datasets. GSE69914 and GSE225845 show a single broad peak of global methylation, whereas GSE287331 exhibits a narrower, right-shifted distribution with two distinct modes, consistent with missing Type I/II probe-bias correction.

To confirm this hypothesis, I stratified all probes by Infinium design type (Type I vs. Type II) using the official Illumina EPIC v1.0 manifest, obtained from the corresponding Bioconductor annotation package [13]. Using this manifest, the β -value distributions were separated by probe design and inspected through a dedicated diagnostic plot (Fig. 12). According to the expected behaviour described by Teschendorff et al. [14], datasets that have undergone BMIQ (or equivalent) normalization should display nearly overlapping β -value distributions for the two probe types. In contrast, the GSE287331 data displayed a clear and systematic separation between Type I and Type II probe distributions, exactly matching the characteristic *uncorrected* bias pattern. This provides strong visual evidence that probe-design normalization was not performed by the original authors.

To further support this conclusion, I examined the official publication associated with GSE287331 [15]. The paper carefully describes the acquisition, processing, and analytical use of Illumina EPIC β -values, but it does *not* mention BMIQ, SWAN, noob, or any other method aimed at correcting Infinium probe-type bias. The absence of such a statement—combined with the strong visual mismatch between Type I and Type II probe distributions—confirms that no probe-design bias correction was applied before the release of the dataset.

For these reasons, applying a dedicated Infinium Type I/II bias correction step is necessary before performing any comparative analysis.

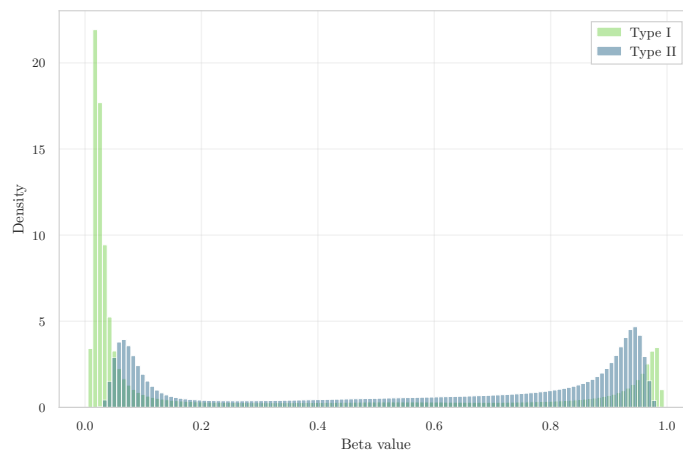


Figure 12: Diagnostic Type I vs. Type II probe design comparison. The strong separation between probe types indicates that no correction was applied.

MANCA LA DESCRIZIONE DELLA CORREZIONE BMIQ
DA QUI IN AVANTI è TUTTO DA RIFARE...

Missingness filtering and imputation Before conducting any exploratory analysis, I quantified the amount and structure of missing values in the raw GSE287331 β -matrix. Probe-wise missingness is a well-known source of unreliability in Illumina methylation arrays, and several studies recommend removing CpGs that exhibit low measurement quality or an excessive proportion of missing values. For example, Islam *et al.* [16] excluded probes with missing β -values in more than 2% of samples in their pediatric tissue methylation analysis, while Mansell *et al.* [17] applied a similar 1% threshold for detection failure in their EPIC-array quality control pipeline.

Guided by this evidence and the large number of CpG sites available in GSE287331 ($> 700,000$ probes), I adopted a stringent probe-wise filtering criterion and removed all CpGs missing in more than 1% of samples. This choice prioritises high-confidence measurements while minimising reliance on imputed values.

After filtering, only a negligible fraction of values remained missing in the working matrix (58 561 NaN, corresponding to 0.0187% of all entries). Given this extremely low level of sparsity, median-based imputation is both statistically robust and distribution-preserving. Following current recommendations, where missing methylation values are replaced by probe-wise medians to avoid introducing group-specific bias, all residual NaN were imputed using the median β -value of each CpG across samples. This approach is consistent with recent harmonization methods such as mLiftOver, where missing methylation values are explicitly replaced using probe-wise median substitution to preserve biologically plausible levels [18].

Global methylation distributions To characterise the overall methylation landscape of the dataset, I first examined the distribution of β -values (fractional methylation in $[0, 1]$). The group-wise mean density curve (Fig. 13) displays the characteristic *bimodal* profile of Illumina 450k arrays, with peaks near unmethylated ($\beta \approx 0$) and fully methylated ($\beta \approx 1$) CpG sites. This pattern reflects the underlying biology of CpG regulation, where many loci tend to be either transcriptionally active (hypomethylated) or repressed (hypermethylated) [2], [5].

When comparing tissue groups, a clear gradient emerges. Tumor samples show a slightly flatter high- β peak and a broader low- β tail, consistent with the well-described phenomenon of **global hypomethylation** and increased heterogeneity in cancer [1]. Tumor-adjacent samples lie between Normal and Tumor, suggesting early epigenetic drift and subtle field effects occurring in histologically non-neoplastic tissue [3]. Normal samples exhibit the sharpest and most stable bimodality, representing the expected physiological baseline.

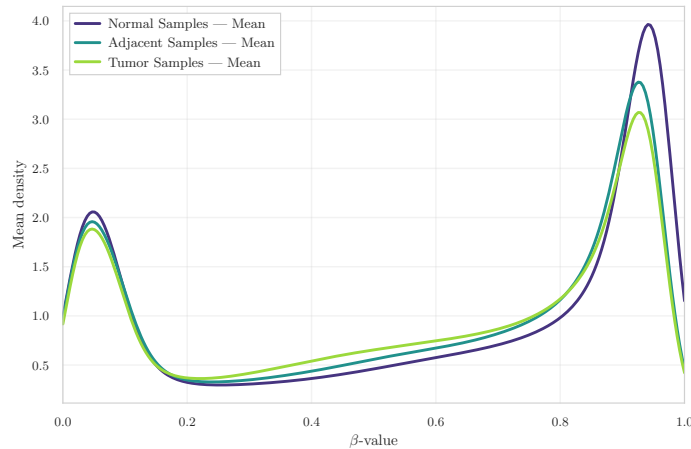
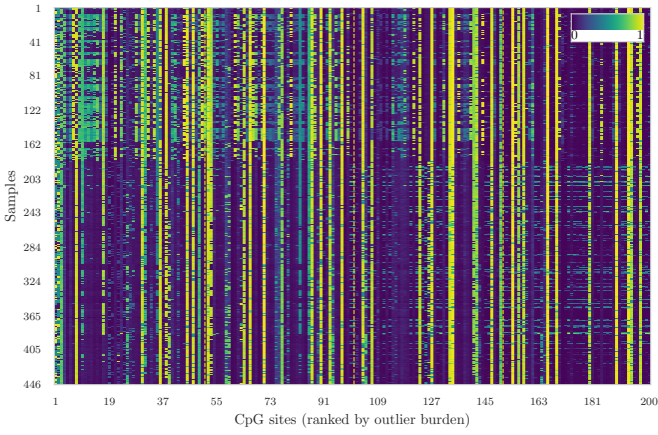


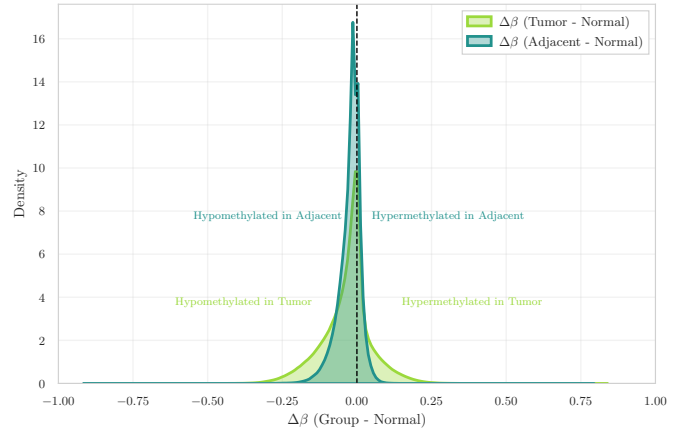
Figure 13: Group-wise mean β -value density.

CpG-level instability and recurrent outliers To characterise locus-specific instability, I examined CpG sites that show the strongest deviations across samples. The heatmap of the top outlier loci (Fig. 14a) indicates that epigenetic disruption is *not uniform*: specific CpG sites and specific samples display extreme deviations, rather than a diffuse genome-wide shift. Moreover, both directions of alteration are present — focal **hypermethylation** (very high β) and focal **hypomethylation** (very low β). In cancer biology, hypermethylation can silence tumor-suppressor regions, whereas hypomethylation can derepress oncogenic pathways and weaken genomic stability, reflecting the classic “too much and too little methylation” behaviour of tumor genomes [1].

Complementary $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Fig. 14b) show a clear shift toward hypomethylation in Tumor samples and a subtler but detectable drift in Adjacent tissues. This provides evidence that epigenetic alterations emerge early in histologically normal tissue located near the tumor, supporting the field-cancerization model [3].



(a) Top outlier CpG loci.

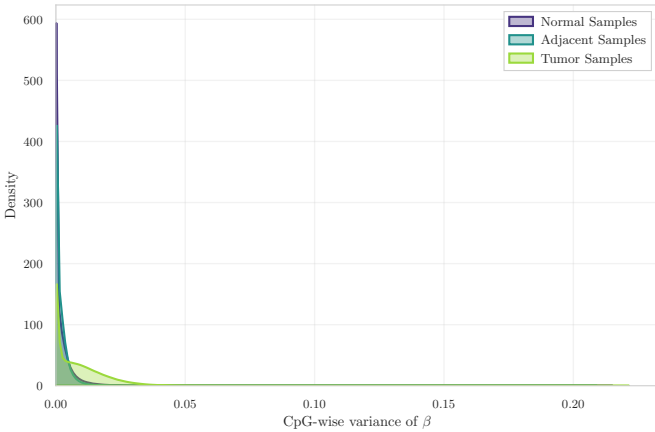


(b) $\Delta\beta$ distributions (Tumor/Adjacent – Normal).

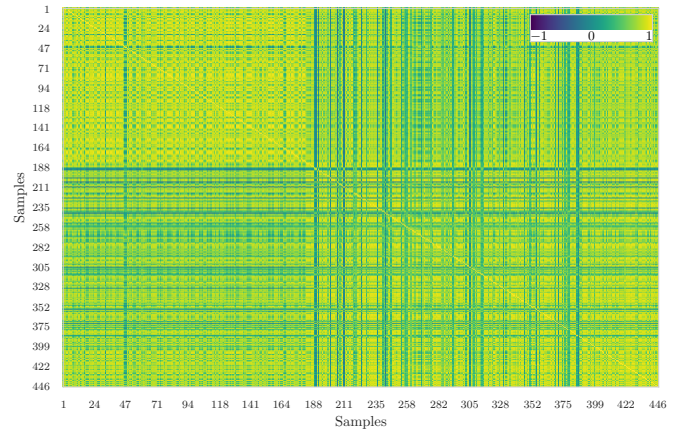
Figure 14: CpG-level instability: heatmap of top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right).

Variance and correlation structure To quantify within-group epigenetic variability, I examined the distribution of CpG-wise variance across samples. The density curves (Fig. 15a) show that Tumor samples have markedly higher variance, reflecting increased epigenetic instability. In contrast, Normal and Tumor-adjacent samples display almost identical variance distributions that are tightly concentrated near zero. This indicates that, at the level of CpG-wise variability, Adjacent tissues do not yet exhibit detectable divergence from Normal samples, despite being histologically proximate to the tumor.

A complementary perspective is provided by the sample correlation heatmap (Fig. 15b), which shows the expected high level of pairwise similarity across all samples. This behaviour is typical of genome-wide DNA methylation data, where a large fraction of CpG sites is stable across individuals, resulting in consistently strong correlations [6].



(a) CpG-wise variance distributions.



(b) Sample correlation heatmap.

Figure 15: Variance and correlation structure across Normal, Adjacent, and Tumor samples.

Low-dimensional embeddings Dimensionality reduction was applied using PCA, t-SNE, and UMAP to visualise global similarities among samples (Fig. 16). Across all three methods, Normal and Tumor-adjacent samples show substantial overlap, with no clear separation between these two groups. Tumor samples appear more dispersed, but only mildly shifted relative to the Normal/Adjacent cluster, and no sharp cluster boundaries emerge in any embedding.

Overall, these projections indicate that global methylation patterns alone do not provide strong discriminative structure among the three tissue states. This suggests that group differences are subtle at the whole-methylome level and are better captured through locus-specific analyses (e.g., outlier burden or $\Delta\beta$), rather than through global unsupervised embeddings.

Genome annotation coverage CpG probes were annotated using the official *Infinium MethylationEPIC v1.0 B5* [7] manifest file, the standard Illumina resource containing genomic coordinates, probe type, CpG island context (Island, Shore, Shelf, Open Sea), and gene-level annotations. Using this manifest, the GSE287331 dataset shows the expected

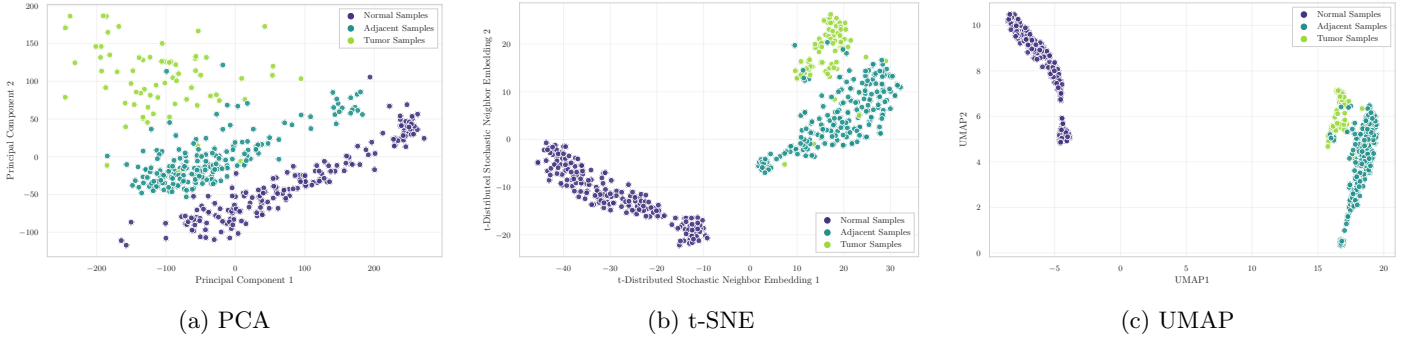


Figure 16: Low-dimensional embeddings of GSE287331 samples.

non-uniform genomic distribution of CpG contexts: the largest fraction of probes maps to open-sea or non-island regions, followed by CpG islands, then north and south shores, and finally north and south shelves. This ordering mirrors the characteristic architecture of the EPIC/HM450 arrays reported in the literature. Zhou et al. [8] describe the EPIC design and its inheritance of the HM450 probe structure, while Pidsley et al. [9] document the enrichment of probes in CpG islands and shores with decreasing density in shelves and open-sea regions. The coverage observed in GSE287331 is therefore consistent with the known genomic composition of the Illumina methylation arrays, supporting correct manifest alignment.

Dynamic-network proxies Although this analysis does not follow the full mathematical formalism of Dynamic Network Biomarkers (DNBs), the degree-based and network-level representations in Fig. 17 provide a qualitative view of how methylation perturbations may organize at the network level. DNB theory, originally formulated for gene-expression and regulatory networks, predicts that near a critical transition a subset of components displays increased internal fluctuations and coordinated behaviour [10], [11], [12]. Here I adapt this conceptual framework to CpG-level methylation patterns by examining the structure of CpG-gene degree distributions and the organization of $\Delta\beta$ -driven network layouts. This allows me to explore whether tumour-associated instability manifests as locally coordinated methylation shifts at the network level, even without implementing the complete DNB pipeline.

The degree-rank plot (Fig. 17a) displays a heavy-tailed distribution of CpG counts per gene, consistent with the heterogeneous hub-periphery structure typically observed in biological systems. In the Tumor vs Normal comparison (Fig. 17b), a compact nucleus of nodes with moderately high degree emerges, accompanied by larger $\Delta\beta$ deviations. This pattern suggests a more perturbed and locally coordinated subset of CpG/gene units in tumor tissue, qualitatively consistent with increased network instability. In contrast, the Adjacent vs Normal network (Fig. 17c) lacks a comparable hub-like structure and shows smaller, more diffuse deviations, indicating that the adjacent tissue may exhibit only early or weakly organized perturbations.

Although these observations do not satisfy the full mathematical criteria for identifying a DNB module, they align conceptually with the idea that tumor tissue occupies a more unstable network regime, whereas adjacent tissue remains closer to the stable (pre-transition) configuration.

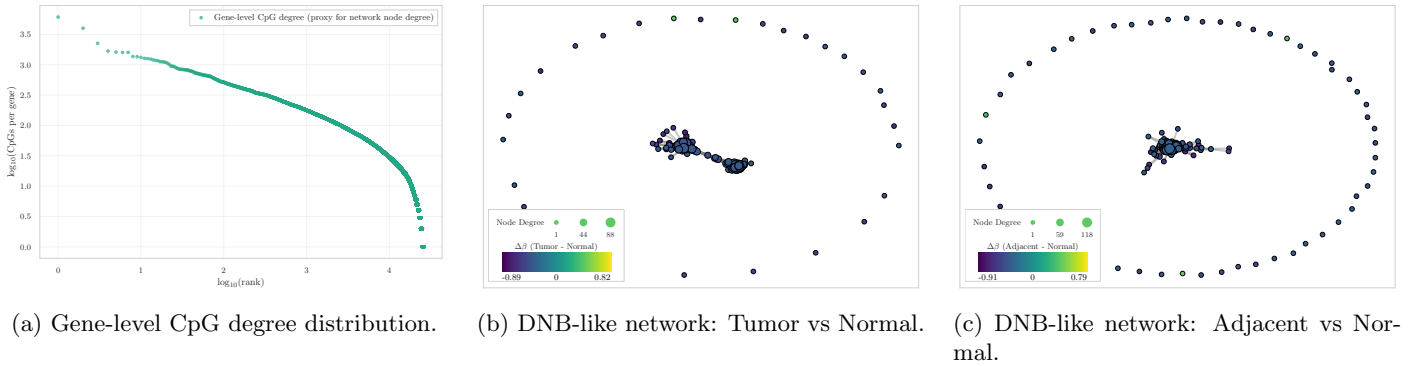


Figure 17: Dynamic-network proxy visualizations for GSE287331.

Summary The exploratory analysis of GSE287331 reveals a consistent but overall subtle separation between the three tissue states. Normal and Tumor-adjacent samples share highly similar global methylation profiles, variance distribu-

tions, correlation structure, and low-dimensional embeddings, indicating that large-scale methylome organisation remains largely conserved in adjacent tissue. Tumor samples show the expected increase in heterogeneity and a shift toward hypomethylation, but these differences remain modest at the whole-methylome level and do not generate clear separation in unsupervised embeddings.

Locus-specific analyses, however, highlight focused patterns of disruption. Outlier CpGs show both hyper- and hypomethylation events, and $\Delta\beta$ distributions reveal a distinct hypomethylation bias in Tumor samples and a milder drift in Adjacent tissue. Dynamic-network proxies further suggest a more compact and perturbed core in Tumor samples, whereas Adjacent tissue remains more diffuse and weakly perturbed.

Altogether, GSE287331 presents a technically clean dataset with well-structured methylation patterns and biologically interpretable deviations. The results indicate that tumour-related alterations emerge predominantly at focal CpG loci rather than at the global methylome level, and that Adjacent tissue reflects early but subtle perturbations consistent with field-effect phenomena.

2 Inter Dataset Analysis

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